

KisanLogic AI - PunjabiMandi.AI

Executive Architecture, Research Innovations & Core Uniqueness Dossier

1. KEY FEATURES & SYSTEM MODULES

- 1. Grassroots Trilingual Localization Engine**
 - Instant context-based toggle across English, Hindi, and Punjabi (Gurmukhi).
- 2. Ground-Reality APMC Profit Simulator**
 - Live moisture (Nami %) penalty deduction matrix and Arhti debt-trap contingency models.
- 3. Game-Theoretic Fair Freight Sharing**
 - Shapley Value cooperative game theory algorithm for fair shared transport cost distribution.
- 4. Inter-Mandi Spatial Arbitrage Optimizer**
 - Distance-matrix net revenue calculations after factoring round-trip diesel, toll, and freight.
- 5. AI Grain Quality & FCI Grading Shield**
 - Physical attribute compliance evaluator generating tamper-proof dispute defense certificates.
- 6. Biological Respiration & Decay Curve**
 - ASABE-standard non-linear daily viability loss curve and critical spoilage cliff projections.
- 7. Circular Stubble-to-Biofuel Valuation Engine**
 - Dynamic biomass yield calculators converting crop residue into guaranteed off-taker revenue.
- 8. Real-Time NASA FIRMS Satellite Tracker**
 - Live MODIS/VIIRS telemetry stream for district-level agricultural fire & stubble monitoring.
- 9. WDRA Smart Cold Storage & e-NWR Hubs**
 - Regional occupancy analytics with direct 7% government pledge loan eligibility markers.
- 10. Multilingual Voice IVR & Zero-Friction Dispatch**
 - Browser telephony voice synthesizer coupled with one-click WhatsApp and PDF reporting.

2. CORE UNIQUENESS & COMPETITIVE MOATS

- **Beyond Price Discovery:**
Bridges the gap between paper mandi rates and actual in-hand realization by subtracting ground deductions (moisture, debt, logistics).
- **Positive Climate Economics:**
Replaces stubble-burning penalties with profitable biomass supply chain monetization.
- **Farmer Bargaining Power:**
Combines cooperative freight game theory with formal FCI grading defense against middle-man exploitation.
- **True Multimodal Accessibility:**
Operable across local vernacular interfaces, automated voice synthesis, and low-bandwidth WhatsApp formats.